RAG Learning

Edexcel A (Salters-Nuffield) A-Level Biology — Core Practicals

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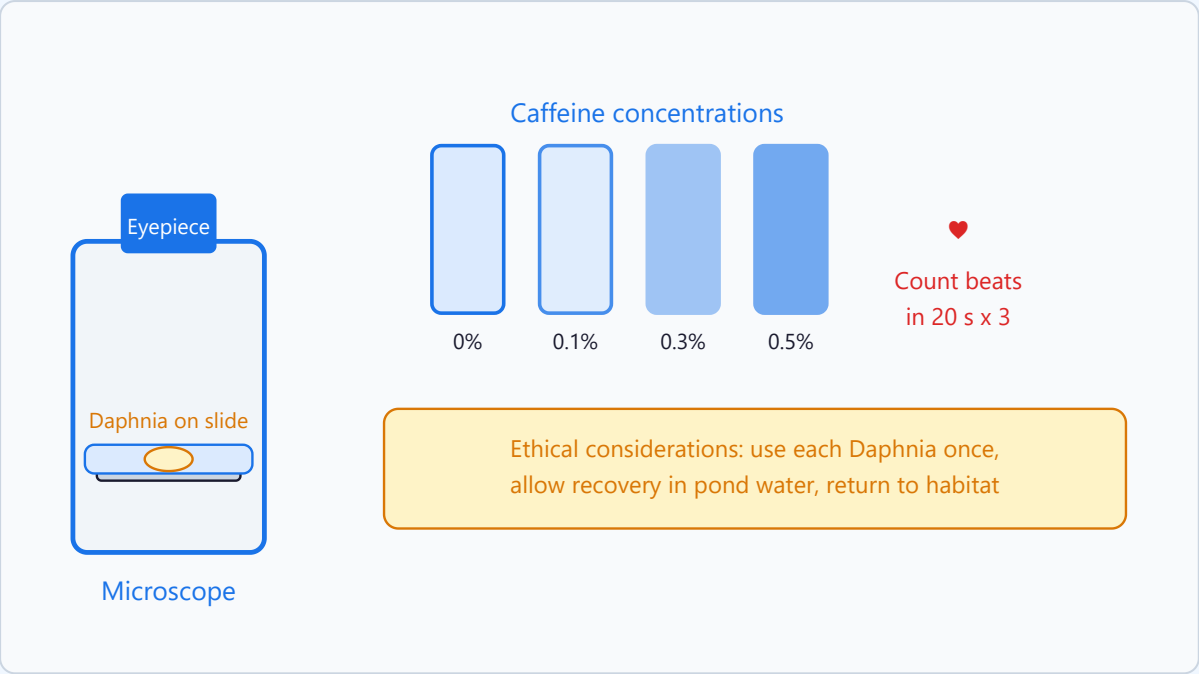
CP1: Effect of Caffeine on Daphnia Heart Rate



Background Theory

Daphnia (water fleas) are small crustaceans with transparent bodies, making their heart visible under a microscope without dissection. Their heart is myogenic (contracts without nervous stimulation) and typically beats at 200–300 bpm at room temperature. **Caffeine** is a stimulant drug that inhibits **phosphodiesterase**, the enzyme that breaks down cAMP (cyclic AMP). This causes cAMP to accumulate, which keeps protein kinase A active for longer, increasing the rate and force of heart contractions. This is the same mechanism by which caffeine increases heart rate in humans.

Apparatus

- Live *Daphnia* (water fleas), cavity slides, cotton wool strands (to restrict movement without crushing)
- Microscope (low power only — high power concentrates too much light/heat), dropping pipette
- Caffeine solutions at a range of concentrations (e.g. 0%, 0.1%, 0.3%, 0.5% w/v), prepared by serial dilution
- Pond water or dechlorinated tap water (for control and recovery), stopwatch

Method

1. Transfer a single *Daphnia* using a wide-bore pipette onto a cavity slide in a small drop of pond water. Place a few cotton wool strands over it to restrict movement without applying pressure.

2. View under low-power microscope. Locate the heart (visible as a pulsating structure in the dorsal part of the transparent body). Allow 2 minutes for the organism to acclimatise.
3. Count heartbeats for **20 seconds**. Multiply by 3 to obtain beats per minute (bpm). Repeat the 20 s count three times and calculate the mean. Using 20 s intervals rather than 60 s reduces the time the Daphnia spends on the slide.
4. Add 2–3 drops of the lowest caffeine concentration to the slide using a pipette. Wait **2 minutes** for the caffeine to diffuse into the organism and take effect.
5. Count heartbeats again (three 20 s counts). Record and calculate mean bpm.
6. Use a **fresh Daphnia** for each concentration to avoid cumulative drug effects. Return all Daphnia to pond water promptly after use.

Key Variables

Variable	Detail
Independent	Caffeine concentration (0%, 0.1%, 0.3%, 0.5%)
Dependent	Heart rate (beats per minute)
Control variables	Temperature (use LED light to minimise heating), acclimatisation time, volume of solution added, size of Daphnia (select similar-sized individuals), same microscope light intensity

Safety and Ethical Considerations

- **Ethical treatment:** Use the minimum number of organisms necessary; handle gently with wide-bore pipettes; minimise time on the slide; return to pond water immediately after experiment; use low concentrations of caffeine that are sub-lethal
- Use **LED or fibre-optic illumination** (not incandescent bulbs) to reduce heat transfer to the Daphnia. A traditional microscope lamp can raise the temperature of the slide by several degrees, independently increasing heart rate
- Each Daphnia should only be used **once** to avoid cumulative/sensitisation effects and ethical concerns about repeated stress
- Caffeine solutions at these concentrations are low hazard to humans

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Analysis

- Calculate mean heart rate (bpm) for each caffeine concentration. Include standard deviation or range as a measure of spread.
- Plot a graph of mean heart rate (y-axis) against caffeine concentration (x-axis). Expect a positive correlation: higher caffeine concentration → higher heart rate.
- Explain using pharmacology: caffeine inhibits phosphodiesterase → cAMP not broken down → cAMP-dependent protein kinase remains active → increased rate of SA node depolarisation → increased heart rate.
- Statistical test: use a Spearman's rank correlation (if testing for correlation) or a Mann-Whitney U test (if comparing two specific groups).

Sources of Error and Improvements

Error	Effect	Improvement
Heart beats too fast to count accurately by eye	Likely underestimate; inconsistency between observers	Use the "tapping" method: tap a pencil on paper in time with heartbeats, then count marks. Alternatively, record video through the microscope and count in slow-motion playback
Heat from microscope lamp	Raises temperature, independently increasing heart rate (confounding variable)	Use LED or fibre-optic light source; limit time under lamp; place a heat filter (beaker of water) between light and stage
Stress from handling and being on a slide	Elevated baseline heart rate due to stress response	Allow 2–3 minutes acclimatisation before first count; handle gently; use wide-bore pipette
Same Daphnia reused for multiple concentrations	Cumulative effect of caffeine; no washout period; ethical concerns	Use a fresh Daphnia for each concentration; randomise order of testing

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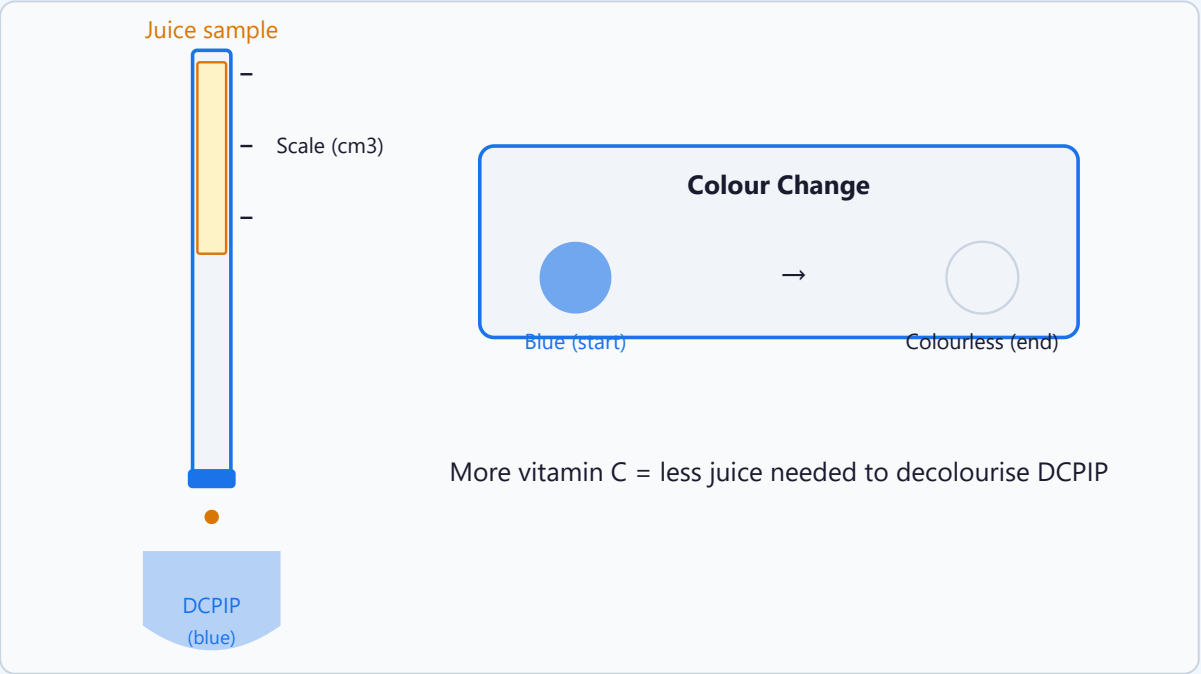
Caffeine concentration may not be uniform on slide	Daphnia may not be exposed to the intended concentration	Gently mix after adding caffeine drops; allow sufficient diffusion time

Exam Tip: This practical is commonly used to test ethical considerations in animal experiments. Always discuss: (1) using the minimum number of organisms, (2) using the lowest effective concentrations, (3) providing recovery time, (4) returning organisms to their habitat, (5) why Daphnia are an appropriate model (transparent body, visible heart, invertebrate so fewer ethical restrictions than vertebrates). Also be prepared to explain why a fresh Daphnia is needed for each concentration — to avoid cumulative drug effects that would confound results.

CP2: Vitamin C Content (DCPIP Titration)



Background Theory

DCPIP (2,6-dichlorophenolindophenol) is a blue dye that acts as an indicator for reducing agents. **Vitamin C (ascorbic acid)** is a powerful reducing agent (antioxidant) that donates electrons to DCPIP, reducing it from its oxidised blue form to its reduced colourless form. The volume of juice required to completely decolourise a fixed volume of DCPIP is inversely proportional to the vitamin C concentration — a higher vitamin C content means less juice is needed.

Apparatus

- DCPIP solution (0.1%), test tubes, test tube rack
- Fruit juice samples (e.g. fresh orange, pasteurised orange, lemon, apple, blackcurrant)
- Graduated pipette (1 cm^3) or burette for accurate volume measurement, syringe
- Known concentrations of ascorbic acid solution (e.g. 0.02%, 0.04%, 0.06%, 0.08%, 0.1% w/v) for constructing a calibration curve
- White tile (to judge colour change against a white background)

Method

1. Pipette exactly **1 cm^3** of DCPIP solution into a test tube.
2. Using a graduated pipette or burette, add the juice sample **drop by drop**, shaking/swirling gently after each addition. Place the tube against a white tile to judge colour.
3. Record the exact volume of juice needed to **completely decolourise** the DCPIP (blue → colourless). This is the end-point.
4. Repeat with each juice sample. Perform at least **3 repeats** per sample and calculate the mean volume.
5. Construct a **calibration curve** by repeating the procedure using known concentrations of ascorbic acid solution instead of juice. Plot volume needed to decolourise (x-axis) against known concentration (y-axis).
6. Use the calibration curve to determine the vitamin C concentration of each unknown juice sample from its mean volume.

Key Variables

Variable	Detail

Independent	Type of juice (or storage method: fresh vs pasteurised vs boiled)
Dependent	Volume of juice needed to decolourise 1 cm^3 DCPIP
Control variables	Volume and concentration of DCPIP, temperature, rate of addition, same batch of DCPIP

Safety

- DCPIP stains skin and clothing — wear gloves and a lab coat.
- Eye protection when handling solutions.
- Ascorbic acid is a mild irritant at high concentrations — low risk at concentrations used.

Analysis

- **Less juice needed** to decolourise = **higher vitamin C content**.
- Plot a calibration curve from known ascorbic acid concentrations. Read off the concentration for each juice from the curve using interpolation.
- Compare different juices or investigate how processing (heating, storage, exposure to air/light) affects vitamin C content.
- Vitamin C is heat-labile and light-sensitive, so pasteurised juice typically has less than fresh juice, and juice stored in clear containers loses more than juice in opaque containers.

Sources of Error and Improvements

Error	Effect	Improvement
Subjective judgement of the colourless end-point	Inconsistent results between experimenters; difficult to judge exact moment of decolourisation	Use a **colorimeter** set to a blue wavelength (~600 nm) to objectively measure absorbance. End-point defined as a specific absorbance value

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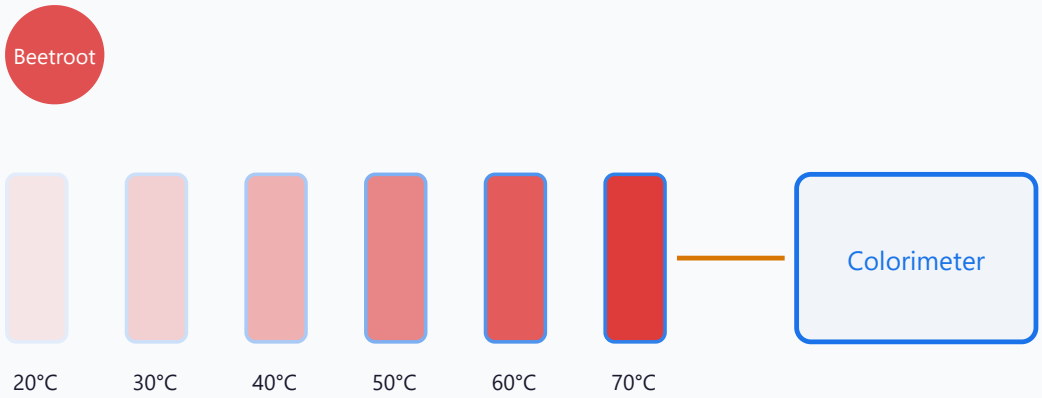
Juice added too quickly (overshooting end-point)	More juice recorded than actually needed	Add drop by drop from a burette or graduated pipette; swirl after each addition; approach end-point slowly
Vitamin C in juice degrades in light and air (oxidation)	Lower apparent vitamin C concentration than the actual original content	Test immediately after opening/preparing; keep samples covered in foil; minimise exposure to air
Pulp or suspended particles in juice	Colour change obscured; difficult to see end-point	Filter juice through muslin or fine mesh before testing
Temperature variation between samples	Rate of reaction between vitamin C and DCPIP varies	Keep all solutions at the same temperature (room temperature)

Exam Tip: Vitamin C (ascorbic acid) is a **reducing agent** that reduces (decolourises) the blue dye DCPIP. More vitamin C means *less* volume of juice needed. Always state the end-point clearly: blue → colourless. If asked about a calibration curve, explain that you plot volume of known ascorbic acid (x) against concentration (y) and use interpolation to find the concentration of unknown samples. Common extension: why does boiled juice need more volume? Because vitamin C is destroyed by heat.

CP3: Membrane Permeability (Beetroot and Temperature)



More pigment leakage = higher absorbance

Background Theory

Beetroot cells contain the red-purple pigment **betacyanin** (a type of betalain) stored in the **vacuole**, enclosed within the tonoplast membrane. Under normal conditions, the intact cell surface membrane and tonoplast are selectively permeable and retain the pigment inside the cell. According to the **fluid mosaic model**, membranes consist of a phospholipid bilayer with embedded proteins. At elevated temperatures: (1) the phospholipid bilayer becomes more fluid as phospholipids gain kinetic energy and move apart, (2) membrane channel and carrier proteins **denature** (their tertiary structure unfolds), creating gaps in the membrane, (3) the membrane loses its selective permeability. This allows betacyanin to leak out of the cell, colouring the surrounding solution. The degree of colour (measured by colorimeter absorbance) is a quantitative measure of membrane damage.

Apparatus

- Fresh beetroot, cork borer (e.g. size 4), scalpel, white tile, ruler
- Test tubes and rack, distilled water (measured volumes)
- Water baths set at 20, 30, 40, 50, 60, 70°C (thermostatically controlled)
- Colorimeter (set to blue-green filter, ~530 nm for betacyanin) and cuvettes, thermometer

Method

1. Use a cork borer to cut **uniform beetroot cylinders**. Trim to identical length (e.g. 1 cm) using a ruler and scalpel on a white tile. This ensures equal surface area for each replicate.
2. Rinse the cylinders **thoroughly** in cold distilled water until the water runs clear. This removes pigment released during cutting from damaged cells, which would give falsely high readings.
3. Place one cylinder in each test tube containing **5 cm^3** distilled water (measure accurately with a measuring cylinder).
4. Place test tubes in water baths at the designated temperatures. Leave for **exactly 20 minutes** (use a stopwatch; stagger starts if necessary).
5. Remove the beetroot cylinder from each tube. Gently shake the solution to ensure even distribution of any leaked pigment.
6. Measure the **absorbance** (or percentage transmission) of each solution using a colorimeter with a blue-green filter (~530 nm). Zero the colorimeter using distilled water as a blank.
7. Repeat the entire procedure at least **3 times per temperature**. Calculate mean absorbance.

Key Variables

Variable	Detail
Independent	Temperature of water bath (20–70°C)
Dependent	Absorbance of the solution (arbitrary units) — a measure of betacyanin leakage
Control variables	Size and surface area of beetroot cylinders, volume of distilled water, time in water bath, same beetroot source, same colorimeter wavelength, thorough initial rinsing

Safety

- Sharp cutting tools — cut on a white tile, always cutting away from the body. Use a cork borer guard if available.
- Hot water baths — use tongs or forceps to handle test tubes in hot water. Take care to avoid burns.
- Beetroot juice stains skin and clothing — wear a lab coat and gloves.

Analysis

- Plot **mean absorbance (y-axis)** against **temperature (x-axis)**.
- Expected result: low, relatively constant absorbance from 20–40°C (membranes mostly intact), followed by a **sharp increase above ~40–50°C** as membrane proteins denature and phospholipid fluidity increases beyond the point of membrane integrity.
- Explain the sigmoid-like curve in terms of the fluid mosaic model: at low temperatures, the membrane remains selectively permeable; above ~40°C, protein denaturation and increased phospholipid movement cause progressive loss of membrane integrity; by 70°C, membranes are essentially destroyed and maximum pigment leakage has occurred.
- Calculate Q_{10} values for different temperature intervals if appropriate.

Sources of Error and Improvements

Error	Effect	Improvement
Unequal cylinder sizes (different surface area:volume ratios)	Different amounts of pigment available for leakage	Use the same cork borer throughout; cut cylinders to the same length with a ruler; weigh cylinders for consistency
Insufficient initial rinsing	Surface pigment from cut cells inflates absorbance readings at all temperatures, including controls	Rinse thoroughly in running distilled water until the rinse water is completely clear
Timing inconsistency between tubes	Some tubes exposed for longer — confounding variable	Use a precise stopwatch; stagger tube placement at 1–2 minute intervals; record exact times
Water bath temperature fluctuations	Actual temperature differs from set temperature	Use thermostatically controlled water baths; check temperature with a thermometer before and after
Air bubbles or debris in cuvette	Inaccurate absorbance readings	Wipe cuvettes with lint-free tissue; check for bubbles; use the same orientation

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Exam Tip: Always explain the increase in permeability in terms of the **fluid mosaic model**: (1) membrane proteins (channel proteins, carrier proteins) denature — their tertiary structure changes, creating gaps, (2) the phospholipid bilayer becomes more fluid as increased kinetic energy disrupts hydrophobic interactions between fatty acid tails, and (3) at very high temperatures, the bilayer may begin to break apart entirely. This allows the red pigment betacyanin to leak out by diffusion from the vacuole. A common exam mistake is saying "the membrane melts" — this is incorrect. The key process is protein **denaturation** and increased **membrane fluidity**.

CP4: Enzyme Activity and Substrate Concentration

Background Theory

This practical investigates how substrate concentration affects the rate of an enzyme-catalysed reaction, typically using **catalase** (which decomposes hydrogen peroxide into water and oxygen: $2H_2O_2 \rightarrow 2H_2O + O_2$). According to the **lock-and-key** and **induced-fit** models, at low substrate concentrations, enzyme active sites are not all occupied, so increasing substrate concentration increases the rate proportionally (first-order kinetics). At high concentrations, **all active sites are saturated** (occupied by substrate), and the rate reaches a maximum (**V_{max}**) that cannot be exceeded without adding more enzyme. This produces the characteristic hyperbolic curve of rate vs substrate concentration.

Apparatus

- Catalase source: uniform potato discs (cut with cork borer) or measured volumes of liver homogenate
- Hydrogen peroxide at various concentrations: 0.5%, 1%, 2%, 4%, 8% (vol/vol), prepared by dilution from stock
- Conical flask, bung with gas delivery tube, gas syringe (or inverted measuring cylinder over water trough)
- Water bath at constant 25°C, stopwatch, measuring cylinder, balance

Method

1. Prepare potato discs of uniform size using a cork borer, then cut to the same mass. Alternatively, use a fixed volume of liver homogenate. Place the enzyme source in a conical flask.
2. Pre-warm the H_2O_2 solution in the water bath for 5 minutes to ensure it is at 25°C.
3. Add the H_2O_2 to the flask and **immediately** fit the bung and start the stopwatch. Speed is critical to capture the initial rate.
4. Record the volume of O_2 collected in the gas syringe every 30 seconds for 5 minutes.
5. Repeat with each H_2O_2 concentration. Perform **3 repeats** per concentration and calculate the mean.
6. To determine the **initial rate** of reaction: draw a tangent to the curve at time = 0 on the volume vs time graph and calculate its gradient.

Safety

- Hydrogen peroxide is an **irritant** and can bleach skin and clothing at higher concentrations — wear eye protection and gloves. Use concentrations ≤8%.
- O_2 supports combustion — no naked flames in the vicinity.
- Take care when fitting the bung tightly — pressure can build up.

Analysis

- Plot **volume of O_2 (y-axis) against time (x-axis)** for each concentration. Calculate initial rate from the gradient of the tangent at t = 0.
- Plot **initial rate (y-axis) against substrate concentration (x-axis)**. Expected: a curve that rises steeply at first (active sites progressively filled), then plateaus at V_{max} (all active sites occupied — enzyme saturation).
- K_m (Michaelis constant) is the substrate concentration at which the rate is half V_{max}. A low K_m indicates high enzyme-substrate affinity.

Sources of Error and Improvements

Error	Effect	Improvement
Gas escapes before the bung is fitted	Volume of O_2 recorded is lower than actual (especially at high concentrations where the reaction is fastest)	Practise fitting the bung rapidly; use a side-arm flask with a rubber septum and inject H_2O_2 through the septum to start the reaction without opening the flask
Variable amount of enzyme between replicates	Unreliable rate comparisons	Use the same mass of potato discs (weigh on a balance) or the same volume of homogenate measured with a syringe
Temperature variation during the experiment	Affects enzyme activity independently of substrate concentration (confounding variable)	Use a thermostatically controlled water bath; monitor temperature throughout
Using "total volume at 5 minutes" instead of initial rate	Substrate depletion means later readings do not reflect the true rate at that concentration	Calculate **initial rate** from the tangent to the curve at t = 0, which reflects the rate when substrate concentration is at its starting value

Exam Tip: At low substrate concentrations, rate increases proportionally because there are many unoccupied active sites (first-order kinetics). At high concentrations, all active sites are occupied (enzyme-substrate complexes forming as fast as they dissociate), so rate plateaus at V_{max} (zero-order kinetics). To increase V_{max}, you must increase the enzyme concentration. Always use **initial rate** for valid comparisons — explain why (substrate concentration decreases over time as it is consumed, so later measurements do not reflect the starting concentration).

CP5: Dissection of Gas Exchange Surfaces

Background Theory

Gas exchange surfaces in organisms are adapted for efficient diffusion of gases (O_2 and CO_2).

According to **Fick's law**: rate of diffusion is proportional to (surface area x concentration gradient) / diffusion distance. All gas exchange surfaces share common adaptations: **large surface area** (maximises area for diffusion), **thin epithelium** (minimises diffusion distance), **rich blood supply** (maintains steep concentration gradient by carrying gases away), and often a **ventilation mechanism** (maintains concentration gradient by replacing air/water).

Apparatus

- Fresh fish head (or whole small fish), dissecting board, pins
- Fine scissors, forceps, scalpel, mounted needle, seeker
- Magnifying glass or dissecting microscope (stereo microscope)
- Gloves, eye protection, lab coat; biological waste bin

Method

1. Place the fish head on the dissecting board. Identify the **operculum** (bony gill cover).
2. Lift the operculum with forceps to expose the **gill arches**. Cut away the operculum carefully with scissors.
3. Identify the **4 gill arches** on each side. Each arch bears many gill filaments (primary lamellae).
4. Carefully remove one gill arch with scissors and forceps. Place it in a Petri dish of water.
5. Examine under magnification. Identify: **gill filaments** (primary lamellae), **secondary lamellae** (plate-like structures on each filament where gas exchange occurs), **gill rakers** (filter food particles), and the rich **blood supply** (afferent and efferent arterioles visible as red/dark structures).
6. Produce a **clear, labelled biological drawing** (pencil, no shading, proportionate) with annotations linking structure to function. Include a scale bar or magnification.

Safety

- Wear gloves when handling biological specimens (risk of bacterial contamination).
- Use sharp instruments carefully; always cut away from your body.
- Dispose of biological tissue in designated biohazard waste containers.
- Wash hands thoroughly after the practical.

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Analysis — Structure-Function Relationships

Structure	Adaptation for Gas Exchange
Many gill filaments with thousands of secondary lamellae	**Large surface area** for gas exchange (Fick's law: rate proportional to area)
Secondary lamellae are very thin (often just 1–2 cells thick)	**Short diffusion distance** between water and blood (Fick's law: rate inversely proportional to distance)
Dense capillary network within each lamella	**Rich blood supply** maintains a steep concentration gradient by rapidly transporting O_2 away and CO_2 towards the exchange surface
Counter-current flow: blood flows in the opposite direction to water flowing over the lamellae	**Maintains a concentration gradient** along the entire length of the lamella, achieving up to 80% O_2 extraction (compared to ~25% if concurrent)
Gill rakers	Filter food particles and debris from water, protecting the delicate lamellae

Sources of Error and Improvements

Error	Effect	Improvement
Structures damaged during dissection	Difficult to identify fine features, especially lamellae	Use fine scissors and gentle technique; cut slowly; practise on less critical structures first
Tissue dries out during examination	Structures shrink and collapse; lamellae stick together	Keep the specimen moist by adding drops of water from a pipette; keep in water when not examining
Drawing not to scale or poorly labelled	Loses marks in assessment	Follow biological drawing conventions: use pencil, clear continuous lines (no sketching), no shading, accurate proportions, labels with ruled lines, include magnification

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Exam Tip: In any dissection or gas exchange question, always apply **Fick's law** and link each structural feature to its function. For the highest marks, explain **counter-current flow**: because blood and water flow in opposite directions, a concentration gradient for O_2 is maintained along the entire length of each lamella, resulting in much more efficient gas exchange than if they flowed in the same direction (concurrent flow). Draw a diagram showing the parallel gradients if asked.

CP6: Rate of Water Uptake (Potometer)

Background Theory

A **potometer** measures the rate of **water uptake** by a leafy plant shoot, which approximates the rate of transpiration (since ~99% of water taken up is lost through transpiration via stomata). Transpiration is the evaporation of water from the surfaces of mesophyll cells inside the leaf, followed by diffusion of water vapour out through open stomata. The rate of transpiration is affected by **environmental factors**: light intensity (opens stomata), temperature (increases kinetic energy of water molecules), wind speed (removes humid air from around stomata, maintaining a steep diffusion gradient), and humidity (reduces the water potential gradient between leaf air spaces and atmosphere).

Apparatus

- Potometer (bubble potometer with calibrated capillary tube), leafy plant shoot (e.g. laurel, privet)
- Rubber tubing and connectors, clamp stand, Vaseline (for sealing joints)
- Water reservoir with tap (to reset the air bubble), stopwatch, ruler
- Fan (to simulate wind), bench lamp (to vary light intensity), plastic bags (to increase humidity), thermometer

Method

1. Cut the plant shoot **underwater** at an angle using sharp secateurs or a razor blade. This prevents air entering the xylem vessels, which would create an airlock and block water flow. Assemble the entire potometer underwater or under running water.
2. Seal all joints with Vaseline to prevent water leakage or air entry. Ensure an airtight seal around the shoot.
3. Allow the apparatus to **equilibrate for 5 minutes** in standard conditions before taking measurements.
4. Introduce an air bubble into the capillary tube by briefly lifting its end out of water, then replacing it. The bubble will move along the scale as the shoot takes up water.
5. Record the **distance the air bubble moves** along the calibrated capillary tube every minute for 5–10 minutes. Calculate the rate of water uptake (mm/min or cm^3/min if the capillary bore is known).
6. Repeat under different conditions: with a fan (increased air movement), with a desk lamp at various distances (increased light intensity), with a plastic bag over the leaves (increased humidity), or at different temperatures.
7. Reset the bubble between readings using the reservoir tap. Allow equilibration before each new measurement.

Key Variables

Variable	Detail
Independent	Environmental condition being tested (e.g. wind speed, light intensity, humidity)
Dependent	Rate of water uptake (distance moved by bubble per unit time)
Control variables	Same shoot (between conditions) or same species/leaf area, temperature (when not being tested), time for equilibration, light conditions (when not being tested)

Safety

- Use care with sharp cutting tools when preparing the shoot (secateurs, razor blades).
- Ensure electrical equipment (fan, lamp) is kept away from water to prevent electric shock.

Analysis

- Rate of water uptake = distance moved by bubble (mm) / time (min). If the internal diameter of the capillary is known, calculate volume: $V = \pi r^2 \times$ distance.
- Expected results: increased wind speed, light intensity, and temperature all **increase** the rate; increased humidity **decreases** the rate.
- Explain each: wind removes the humid boundary layer around stomata (steeper diffusion gradient); light opens stomata (K^+ pumps in guard cells cause water uptake by osmosis, making guard cells turgid and opening the pore); heat increases kinetic energy of water molecules (faster evaporation); humidity reduces the water potential gradient between leaf interior and atmosphere.

Sources of Error and Improvements

Error	Effect	Improvement
Air lock in xylem vessels	Water cannot flow; bubble does not move; apparatus fails	Always cut the shoot **underwater**; assemble apparatus underwater; check for air bubbles in tubing
Leaks at joints or around the shoot	Water loss not due to transpiration; results invalid	Apply Vaseline to all connections; test for leaks before starting
Temperature changes affecting gas expansion	Air bubble moves due to thermal expansion, not water uptake	Control temperature; equilibrate fully; use a water jacket around the capillary tube
Leaves wilting during long experiments	Stomata close; transpiration rate drops; not representative	Keep experiments short; use well-watered shoots; reset frequently

Exam Tip: A potometer measures **water uptake**, not transpiration directly. A small proportion of water taken up is used for photosynthesis, cell expansion, and maintaining turgor. However, since ~99% is lost through transpiration, water uptake is a very close approximation. Always state this distinction in exam answers. If asked to compare conditions, explain the mechanism: e.g. "wind increases transpiration because it removes the layer of

humid air from around the stomata (boundary layer), maintaining a steep water potential gradient between the saturated air spaces inside the leaf and the drier air outside."

CP7: Antimicrobial Substances (Zones of Inhibition)

Background Theory

This practical tests the effectiveness of different antimicrobial agents (antibiotics, antiseptics, or natural plant extracts) against bacteria. When antimicrobial discs are placed on an agar plate seeded with bacteria, the agent **diffuses outward** through the agar, creating a concentration gradient. Where the concentration exceeds the minimum inhibitory concentration (MIC), bacteria are killed or inhibited, producing a clear zone called the **zone of inhibition**. A larger zone indicates a more effective antimicrobial agent (or one that diffuses more readily through the agar). This is the **Kirby-Bauer disc diffusion method**, widely used in clinical microbiology to guide antibiotic selection.

Apparatus

- Pre-poured nutrient agar plates, non-pathogenic bacterial broth culture (e.g. *E. coli* K-12 or *Micrococcus luteus*)
- Antibiotic multidiscs (or individual discs soaked in different antibiotics/plant extracts)
- Sterile glass spreader (or sterile cotton swab for streak inoculation), Bunsen burner, forceps (flamed for sterilisation)
- Marker pen, adhesive tape (2 pieces only — not fully sealed), ruler (mm scale), incubator

Aseptic Technique (Essential for Validity and Safety)

- Work near a **lit Bunsen burner** — the updraft of hot air creates a sterile zone by carrying airborne microorganisms upward and away from the work area
- **Flame** the neck of bottles and culture tubes before and after opening (kills organisms on the rim)

- Sterilise inoculating loops and spreaders by **flaming until red-hot**; allow to cool before use (or use pre-sterilised disposable plastic loops)
- Keep lids on Petri dishes as much as possible; do not breathe, cough, or talk over open plates
- Secure lids with **two pieces** of adhesive tape (not fully sealed — anaerobic conditions could promote growth of dangerous obligate anaerobes like *Clostridium*)

Method

1. Working near a Bunsen flame, use a sterile cotton swab or glass spreader to **spread the bacterial culture evenly** across the entire surface of the agar plate. This creates a uniform "lawn" of bacteria.
2. Using **flamed forceps**, place the antibiotic discs (or multidisc) on the agar surface. Press gently to ensure contact. Space discs evenly (at least 25 mm apart to prevent overlapping zones).
3. Seal the lid with two pieces of tape (forming an X). **Do not fully seal**. Label the base of the plate with name, date, organism, and disc identities.
4. Incubate **upside down** (prevents condensation dripping onto colonies) at **≤25°C** for 24–48 hours.
5. After incubation, **do not open the plate**. Measure the **diameter** of each zone of inhibition in mm using a ruler held against the base of the plate. Measure at the widest point.

Safety

- Maximum incubation temperature in schools: **≤25°C**. At 37°C (human body temperature), dangerous human pathogens could proliferate if contamination occurred.
- **Never open plates after incubation**. Unknown organisms may have grown.
- After examination, plates must be **autoclaved** (121°C, 15 min, 15 psi) before disposal in clinical waste.
- Full aseptic technique throughout to prevent contamination (both of the experiment and of the experimenter).

Analysis

- Calculate the **area** of each zone of inhibition: $A = \pi r^2$, where r = diameter/2. Area is more appropriate than diameter because the antimicrobial diffuses radially in two dimensions.
- Subtract the disc area if required (for a more accurate representation of the inhibited area).

- A larger zone indicates a more effective antimicrobial agent against that bacterium (or better diffusion through the agar).
- Note: zone size depends on both the potency of the agent AND its ability to diffuse through agar. A highly effective agent with poor diffusion may produce a small but very clear zone.

Sources of Error and Improvements

Error	Effect	Improvement
Contamination from the environment	Extra colonies obscure zones; unreliable results	Strict aseptic technique; work near Bunsen flame; sterilise all equipment
Uneven bacterial lawn	Inconsistent zone shapes; difficult to measure accurately	Spread thoroughly and evenly using a sterile spreader; rotate the plate while spreading
Discs not in good contact with agar surface	Poor diffusion of antimicrobial; smaller or absent zones	Press each disc gently but firmly with flamed forceps to ensure full contact
Overlapping zones	Cannot measure individual zones accurately	Space discs at least 25 mm apart; use a standard template
Measuring diameter instead of calculating area	Underestimates the proportional difference between zones (area scales with r^2)	Always calculate area using $A = \pi r^2$

Exam Tip: Always calculate **πr^2** for the zone area (not just diameter). Larger zone area = more effective antibiotic. Explain why incubation is at **≤25°C in schools**: to prevent the growth of potentially dangerous human pathogens (which thrive at 37°C). Explain why plates are incubated **upside down**: to prevent condensation from the lid dripping onto the agar surface, which could cause colonies to merge and spread, making them impossible to count or measure.

CP8: Factors Affecting Photosynthesis Rate

Background Theory

Photosynthesis rate is influenced by three main **limiting factors**: light intensity, CO_2 concentration, and temperature. At any given time, the factor in shortest supply limits the overall rate (Blackman's law of limiting factors). This practical typically investigates **light intensity** using aquatic pondweed (*Elodea* or *Cabomba*), which produces visible O_2 bubbles during photosynthesis. Light intensity is inversely proportional to the square of the distance from the light source: $\mathbf{I \propto 1/d^2}$ (the inverse square law).

Apparatus

- *Elodea* (Canadian pondweed) or *Cabomba*, beaker, sodium hydrogen carbonate solution ($NaHCO_3$, ~1%)
- Bench lamp (with known wattage), ruler (to measure distance), clamp stand
- Stopwatch, thermometer, heat shield (beaker of water between lamp and plant)
- Optional: gas syringe or photosynthometer for more accurate volumetric O_2 measurement

Method

1. Cut a healthy sprig of *Elodea* at an angle (to maximise the cut surface for bubble release) and place it **upside down** in a beaker of $NaHCO_3$ solution. The $NaHCO_3$ provides a constant supply of dissolved CO_2, ensuring CO_2 is not a limiting factor.
2. Position the bench lamp at a set distance from the beaker (e.g. 5 cm). Place a heat shield (large beaker of water) between the lamp and the plant beaker to absorb infrared radiation and prevent temperature changes.
3. Allow **2 minutes** for the plant to equilibrate at each new light intensity. Discard any bubbles produced during this time (the plant needs time to adjust its photosynthetic rate).
4. Count the number of **O_2 bubbles** released per minute. Repeat the count **3 times** at each distance and calculate the mean rate.
5. Move the lamp to different distances (e.g. 5, 10, 15, 20, 25, 30 cm). Record bubble count at each distance.

6. Calculate relative light intensity at each distance: **$I = 1/d^2$** (arbitrary units).

Safety

- Keep the electrical lamp away from water — electrical shock hazard.
- The lamp gets hot — do not touch the bulb. Use the heat shield to prevent heating the pondweed.
- $NaHCO_3$ is an irritant in concentrated solutions — wash hands if contact occurs.

Analysis

- Plot **mean bubble rate (y-axis)** against **$1/d^2$** (light intensity, x-axis).
- Expected result: initially, rate increases proportionally with light intensity (light is the limiting factor). At higher intensities, the curve plateaus — another factor (temperature or CO_2) becomes limiting.
- The plateau demonstrates that **light is no longer the limiting factor**. At this point, increasing light intensity has no further effect because the rate is limited by the concentration of CO_2, the availability of RuBisCO, or the temperature-dependent rate of the Calvin cycle.

Sources of Error and Improvements

Error	Effect	Improvement
Bubbles vary in size	Counting bubbles does not accurately represent volume of O_2	Use a **gas syringe or photosynthometer** to measure actual volume of O_2 produced
Heat from lamp raises temperature at close distances	Temperature confounds the effect of light; rate increase may be partly due to heat	Use a **heat shield** (water-filled beaker); monitor temperature with a thermometer; use an LED lamp (less heat)
Background light from room/windows	Additional, uncontrolled photosynthesis occurs at all distances	Work in a **darkened room**; draw blinds; use blackout material

Not all bubbles are pure O_2	Some bubbles may be dissolved air coming out of solution due to heating	Allow the plant to equilibrate; discard the first 2 minutes of data at each distance; use the heat shield
Insufficient equilibration time	Rate has not stabilised; data at each distance is unreliable	Allow at least 2 minutes at each new distance before counting

Exam Tip: Light intensity is inversely proportional to the square of the distance: **$I \propto 1/d^2$**. $NaHCO_3$ is added to provide a **constant supply of CO_2** so that it does not become a limiting factor. Always state both points. When explaining the plateau, say: "At high light intensities, another factor such as CO_2 concentration or temperature becomes the limiting factor, so further increases in light intensity cannot increase the rate of photosynthesis." This is Blackman's law of limiting factors.

CP9: Factors Affecting Respiration Rate (Respirometer)

Background Theory

A **respirometer** measures the rate of aerobic respiration by detecting the uptake of O_2. Germinating seeds are used because they have a high metabolic rate. The key design feature is the inclusion of **soda lime** ($NaOH/Ca(OH)_2$), which absorbs all CO_2 produced by respiration. Since CO_2 is removed, any decrease in gas volume within the closed system is solely due to O_2 consumption, causing a coloured liquid drop in the capillary tube to move towards the organisms. A **glass bead control** tube (same volume as the seeds but with no respiration) accounts for any changes in atmospheric pressure or temperature that might independently affect gas volume.

Apparatus

- Simple respirometer (two boiling tubes or syringes connected to a capillary tube with a manometer fluid drop)
- Germinating seeds (e.g. mung beans, peas — germinated for 2–3 days), glass beads (control)

- Soda lime granules (CO_2 absorbent), cotton wool plugs (prevent soda lime contacting organisms)
- Capillary tube with scale (mm markings), coloured manometer fluid (e.g. potassium permanganate)
- Water bath (thermostatically controlled), thermometer, stopwatch

Method

1. Place a layer of soda lime at the bottom of each tube, separated from the organisms by a cotton wool plug. In one tube, place germinating seeds; in the control tube, place an equal volume of glass beads on top of soda lime (also separated by cotton wool).
2. Seal both tubes and connect them to the capillary tube containing the coloured fluid drop. Place the entire apparatus in a water bath at the desired temperature. **Equilibrate for 5 minutes**.
3. Record the starting position of the coloured fluid drop. Then record its position every minute for 10–15 minutes.
4. The drop moves **towards the organisms** as O_2 is consumed (net decrease in gas volume because CO_2 is absorbed by soda lime).
5. Repeat at different temperatures (e.g. 15, 20, 25, 30, 35, 40°C) to investigate the effect of temperature.

Safety

- **Soda lime is corrosive** (strongly alkaline) — wear chemical-resistant gloves and eye protection. Do not allow skin contact. If contact occurs, wash immediately with copious water.
- Do not allow soda lime to touch the organisms (separated by cotton wool).
- Handle hot water baths carefully — risk of scalding.

Analysis

- Rate of O_2 uptake = distance moved by fluid drop (mm) / time (min).
- If capillary bore diameter is known, convert to volume: $V = \pi r^2 \times$ distance.
- Plot **rate of O_2 uptake against temperature**. Expected: rate increases with temperature (increased kinetic energy of enzyme molecules; more frequent enzyme-substrate collisions),

roughly doubling for every 10°C rise (**Q_{10} ≈ 2**), until an optimum is reached. Beyond ~40°C, rate decreases as respiratory enzymes denature.

- **Q_{10}** = rate at (T + 10) / rate at T. A Q_{10} of ~2 confirms the process is enzyme-catalysed.

Sources of Error and Improvements

Error	Effect	Improvement
Leaky apparatus (poor seals)	Air enters; fluid drop does not move reliably	Seal all joints with Vaseline or silicone grease; check for leaks before starting by watching for unexpected movement
Seeds at different germination stages	Variable metabolic rates between replicates	Use seeds from the same batch, germinated for exactly the same number of days
No control tube (glass beads)	Cannot distinguish O_2 uptake from atmospheric pressure/temperature changes	Always include the glass bead control; subtract control movement from experimental reading
Soda lime exhausted (saturated with CO_2)	CO_2 no longer absorbed; volume change no longer reflects O_2 uptake alone	Use fresh soda lime for each experiment; do not run experiments for excessively long periods
Insufficient equilibration time	Temperature not uniform; initial readings unreliable	Allow at least 5 minutes for temperature equilibration before recording

Exam Tip: Explain why the soda lime is essential: it absorbs CO_2 produced by respiration, so the only gas change is O_2 consumption, causing a net decrease in volume. Without soda lime, the CO_2 produced would partially replace the O_2 consumed (RQ ≈ 1 for glucose), and the fluid drop would barely move. The glass bead control accounts for changes in atmospheric

pressure and temperature. State Q_{10} = rate at (T+10) / rate at T. If $Q_{10} \approx 2$, the process is enzyme-catalysed.

CP10: Effects of Exercise on Ventilation

Background Theory

During exercise, muscles contract more frequently, increasing the rate of aerobic respiration: glucose + $O_2 \rightarrow CO_2 + H_2O$ + ATP. This elevates blood CO_2 concentration and lowers blood pH ($CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$). **Central chemoreceptors** in the medulla oblongata and **peripheral chemoreceptors** in the carotid and aortic bodies detect the rise in CO_2/H^+ and send nerve impulses to the respiratory centre, which increases both **breathing rate** (breaths per minute) and **tidal volume** (volume per breath). **Minute ventilation** = tidal volume x breathing rate, and this increases to supply more O_2 and remove more CO_2.

Apparatus

- Spirometer or peak flow meter, clean disposable mouthpieces (one per participant)
- Stopwatch, step bench or exercise bike, metronome (to standardise step rate)
- Nose clip (essential for spirometry to ensure all air passes through the mouthpiece)
- Questionnaire for participant screening (asthma, heart conditions, recent illness)

Method

1. **Screen participants** for respiratory or cardiac conditions before beginning. Obtain informed consent.
2. Measure **resting breathing rate** (count breaths per minute for 60 s) and **resting tidal volume** (using spirometer with nose clip).
3. Subject performs standardised exercise (e.g. step-ups at 30 steps per minute for 5 minutes, using a metronome for consistency).
4. **Immediately after** exercise ceases, measure breathing rate and tidal volume again.

5. Continue measuring at **1-minute intervals during recovery** for 5–10 minutes until values return to baseline.
6. Calculate **minute ventilation** = tidal volume x breathing rate at each time point.

Safety

- Screen participants for respiratory conditions (asthma), cardiac problems, and injuries.
- Use clean, individual disposable mouthpieces (hygiene — prevent cross-infection).
- Stop exercise immediately if the participant feels dizzy, faint, experiences chest pain, or feels unwell.
- Have a first-aid kit available; work in a well-ventilated room.

Analysis

- Plot breathing rate and tidal volume as separate graphs against time (before, during, and after exercise).
- Calculate and plot minute ventilation against time.
- Both breathing rate and tidal volume increase during exercise. Recovery time (time to return to resting values) can be measured and compared between individuals.
- Fitter individuals typically have a shorter recovery time, larger tidal volumes, and lower resting heart rates.

Sources of Error and Improvements

Error	Effect	Improvement
Inconsistent exercise intensity between participants	Cannot compare results fairly	Use a metronome to standardise step rate; use the same step height and duration for all
Air escaping through nose during spirometry	Tidal volume measurement underestimated	Use a nose clip; ensure proper mouthpiece seal

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Individual variation in fitness levels	Large between-subject variation makes trends hard to see	Compare within the same individual (before vs after); use large sample sizes; calculate means
Delay between stopping exercise and first measurement	Recovery has already begun; peak values missed	Start measurements immediately (<5 seconds) after exercise stops

Exam Tip: Minute ventilation = tidal volume x breathing rate. During exercise, CO_2 concentration in the blood rises, which is detected by chemoreceptors (central in the medulla, peripheral in the carotid/aortic bodies). These send increased nerve impulses to the respiratory centre in the medulla, which increases the rate and depth of breathing. This is an example of **negative feedback** maintaining blood gas homeostasis.

CP11: Effect of Gibberellin on Seed Germination

Background Theory

Gibberellic acid (GA_3) is a plant growth regulator that plays a crucial role in seed germination. In cereal seeds (e.g. barley, wheat), the germination process involves: (1) water is absorbed by the seed (imbibition), (2) the embryo produces GA_3 and secretes it towards the **aleurone layer** (a protein-rich layer surrounding the starchy endosperm), (3) GA_3 stimulates cells in the aleurone layer to synthesise and secrete **alpha-amylase** (and other hydrolytic enzymes), (4) alpha-amylase diffuses into the endosperm and **hydrolyses starch into maltose**, (5) maltose is transported to the embryo and used as a respiratory substrate to provide energy for growth. Exogenous application of GA_3 to seeds can promote or accelerate germination.

Apparatus

- Seeds (e.g. wheat or barley grains — ideally de-embryonated to demonstrate the role of exogenous GA_3)

- Petri dishes, cotton wool or filter paper
- Gibberellic acid solution at various concentrations (e.g. 0, 10, 100, 1000 ppm)
- Distilled water (for control), ruler, labels, incubator or warm cupboard

Method

1. Place 10 seeds on damp cotton wool or filter paper in each labelled Petri dish.
2. Add equal volumes of distilled water (control) or GA_3 solution at different concentrations.
3. Ensure the cotton wool is moist but not waterlogged (seeds need O_2 for aerobic respiration).
4. Incubate all dishes in the **same conditions** (temperature ~20–25°C, in darkness to eliminate phototropism as a variable).
5. Record **germination percentage** (number germinated out of 10, expressed as %) and **shoot/root length** daily for 7–14 days.
6. Use at least 3 replicates per concentration.

Safety

- GA_3 solutions at the concentrations used are low hazard. Wash hands after handling.
- Avoid inhaling seed dust (may cause allergic reactions in sensitive individuals).
- Dispose of mouldy seeds/plates in biohazard waste.

Analysis

- Plot **germination percentage** and **mean shoot/root length** against time for each GA_3 concentration.
- Higher GA_3 concentrations are expected to increase the rate and percentage of germination (up to an optimum), and may increase early shoot elongation.
- Explain the mechanism: GA_3 binds to a receptor protein in aleurone cells → activates transcription factors → gene expression of alpha-amylase is switched on → amylase secreted into endosperm → starch hydrolysed to maltose → provides respiratory substrate (glucose after further hydrolysis by maltase) for embryo growth.

Sources of Error and Improvements

Error	Effect	Improvement
Variable seed quality/viability within batch	Some seeds may fail to germinate regardless of treatment	Use seeds from the same batch and supplier; pre-test viability of a sub-sample
Unequal volumes of solution	Different water availability between dishes; confounding variable	Measure exact volumes with a graduated pipette or syringe
Temperature variation between dishes	Different germination and growth rates	Keep all dishes in the same incubator or location; monitor temperature
Fungal contamination of seeds	Moulds can inhibit germination or obscure results	Surface-sterilise seeds in dilute bleach (1%) for 2 minutes before use, then rinse

Exam Tip: Gibberellin (GA_3) stimulates the aleurone layer to produce **alpha-amylase**, which hydrolyses starch to maltose. This is a key example of hormonal regulation of gene expression in plants. De-embryonated seeds treated with GA_3 show starch digestion, proving that GA_3 (not the embryo itself) triggers amylase production. This links Topic 8 (gene expression) with practical work.

CP12: Tensile Strength of Plant Fibres

Background Theory

Plant cell walls contain **cellulose microfibrils** — long, unbranched chains of beta-glucose linked by 1,4-glycosidic bonds, held together by hydrogen bonds into bundles. These microfibrils provide exceptional tensile strength (resistance to stretching/breaking). **Sclerenchyma fibres** have additional secondary cell wall thickening and are often **lignified**, making them extremely strong. This practical measures the tensile strength of plant fibres by progressively loading them until they

break, and relates the breaking force to the cross-sectional area of the fibre. Tensile stress = force / cross-sectional area.

Apparatus

- Plant fibres (e.g. celery petiole strings, nettle bast fibres, sclerenchyma strands)
- Clamp stand, boss, clamp (or a purpose-built fibre-testing rig)
- Masses (10 g, 20 g, 50 g increments), hook or S-clip, container (to catch fallen masses)
- Ruler, micrometer (eyepiece graticule and stage micrometer) or digital callipers for measuring fibre diameter

Method

1. Carefully extract a single fibre from the plant using forceps and a mounted needle. Avoid damaging the fibre during extraction.
2. Measure the **diameter** of the fibre at several points along its length using a micrometer or calibrated microscope eyepiece graticule. Calculate the mean diameter and the radius (r). Calculate cross-sectional area: $A = \pi r^2$.
3. Clamp one end of the fibre securely. Attach a hook to the lower end.
4. Add masses one at a time (e.g. 10 g increments). Wait ~10 seconds after each addition for the fibre to stabilise.
5. Record the mass at which the fibre **breaks**.
6. Repeat for at least **10 fibres** per plant species to obtain a reliable mean. Calculate mean breaking mass.
7. Convert mass to force: F = m x g (where g = 9.81 N/kg).
8. Calculate **tensile stress** = F / A (in Pa or MPa).

Safety

- Place a cushion, tray, or container below the hanging masses to catch them if they fall when the fibre breaks.
- Wear eye protection in case the fibre snaps suddenly and recoils.
- Take care when extracting fibres (sharp tools may be needed).

Analysis

- **Force at break** = mass (kg) x 9.81 (N/kg). Convert g to kg first.
- **Cross-sectional area** = πr^2 (assuming a circular cross-section).
- **Tensile stress** = force / area. Report in MPa (megapascals). Higher stress = stronger fibre.
- Compare fibres from different species, or from different parts of the same plant (e.g. stem vs petiole).
- Relate strength to cell wall composition: cellulose microfibril density, lignification, secondary wall thickness.

Sources of Error and Improvements

Error	Effect	Improvement
Fibre damaged during extraction	Weakened structure; breaks at lower force than its true tensile strength	Extract carefully using fine forceps and a mounted needle; avoid bending, kinking, or stretching
Variable fibre diameter along its length	Cross-sectional area calculation inaccurate; stress may be over- or underestimated	Measure diameter at multiple points (e.g. 3–5 positions); calculate mean diameter; acknowledge this as a source of uncertainty
Fibre slips out of clamp instead of breaking	Records a lower apparent breaking force; not a true measure of tensile strength	Wrap the fibre securely around the clamp; use rubber pads to grip; ensure the clamp is tight
Small sample size	Mean may not be representative; high uncertainty	Test at least 10 fibres per type; report standard deviation and range
Fibre cross-section is not circular	$A = \pi r^2$ assumes a circle; actual area may differ	Examine cross-sections under a microscope; use image analysis software for more accurate area measurement

Exam Tip: Tensile strength is primarily due to **cellulose microfibrils** in the cell wall, which run in parallel and are cross-linked by hydrogen bonds. **Sclerenchyma fibres** are particularly

strong because they have thick secondary walls with additional cellulose layers and **lignin** deposits (which also makes them waterproof). Compare with collagen fibres in animals, which provide tensile strength in tendons.

CP13: DNA Extraction from a Plant

Background Theory

DNA can be extracted from cells by exploiting its physical and chemical properties. The process involves three key steps: (1) **cell lysis** — breaking open cell walls and membranes to release the cell contents, (2) **separation** — isolating DNA from other cellular components (proteins, lipids, RNA), and (3) **precipitation** — causing DNA to come out of solution in a visible, spoolable form. Strawberries are commonly used because they are **octoploid (8n)**, meaning each cell contains 8 copies of each chromosome and therefore a large amount of DNA per cell.

Apparatus

- Strawberries (or kiwi fruit, onion, peas), zip-lock bag
- Extraction buffer: ~10 cm^3 washing-up liquid (detergent) + 3 g table salt (NaCl) + 100 cm^3 warm water (~60°C)
- Muslin cloth or tea strainer, beaker, boiling tube or test tube
- Ice-cold ethanol (>95%, kept in a freezer), glass rod (for spooling)

Method

1. Place 2–3 strawberries in a zip-lock bag and mash thoroughly with fingers for **3–5 minutes**. This physically breaks open cell walls (mechanical disruption).
2. Add ~20 cm^3 of the extraction buffer to the bag and mix gently (avoid excessive bubbling from the detergent) for 2 minutes.
3. Filter the mixture through muslin cloth into a beaker. Transfer the clear filtrate to a boiling tube (fill about one-third full).

4. Very slowly and gently, pour **ice-cold ethanol** down the inside wall of the tube so that it forms a distinct layer on top of the filtrate (ethanol is less dense). Do not mix the layers.
5. Wait 2–3 minutes. White, stringy/cloudy material will appear at the interface between the two layers — this is **precipitated DNA**.
6. Use a glass rod to **gently spool the DNA** by slowly twisting the rod at the interface. The DNA wraps around the rod and can be lifted out.

Why Each Reagent Works

Reagent	Function	Biochemical Explanation
Washing-up liquid (detergent)	Lyses cell membranes and nuclear membranes	Detergent molecules are amphipathic (hydrophilic head, hydrophobic tail) — they dissolve the phospholipid bilayer by disrupting hydrophobic interactions between fatty acid tails, releasing the cell contents including DNA
Salt (NaCl)	Causes DNA to clump together; helps separate proteins from DNA	Na^+ ions neutralise the negative charges on the phosphate groups of the DNA backbone. This reduces electrostatic repulsion between DNA strands, allowing them to aggregate. It also causes histone proteins to dissociate from DNA
Warm water (~60°C)	Optimises lysis; helps denature proteins	Heat increases membrane fluidity (aiding lysis) and denatures DNase enzymes that would otherwise degrade the DNA. Not too hot (>80°C) or the DNA itself would denature
Ice-cold ethanol	Precipitates DNA out of solution	DNA is soluble in water but **insoluble in cold ethanol**. When cold ethanol is layered on top, DNA comes out of solution at the aqueous-ethanol interface as white, stringy threads. Cold temperature reduces the activity of any remaining DNases

Safety

- Ethanol is **highly flammable** — no naked flames nearby. Keep the stock bottle closed.

- Detergent solution is an irritant — wash hands if prolonged skin contact occurs.
- Take care with warm water — risk of minor scalding.

Sources of Error and Improvements

Error	Effect	Improvement
Ethanol not cold enough	Less DNA precipitates; yield is lower	Keep ethanol in a freezer (−20°C) for at least 1 hour before use
Mixing the ethanol and filtrate layers vigorously	DNA fragments into small pieces that are impossible to spool	Layer ethanol **very gently** down the side of the tube; do not shake or stir
Insufficient mashing	Fewer cells lysed; lower DNA yield	Mash thoroughly for at least 3–5 minutes; ensure no large chunks remain
Too much detergent (excess foam)	Foam interferes with observation of DNA precipitation	Use measured amount of detergent; mix gently to minimise bubbles

Exam Tip: Know the role of each chemical: **detergent** dissolves phospholipid membranes (disrupts hydrophobic interactions); **salt** neutralises the negative charges on DNA phosphate groups (allows aggregation); **cold ethanol** precipitates DNA (DNA insoluble in ethanol). Strawberries are often chosen because they are **octoploid (8n)**, so contain lots of DNA per cell. The extracted DNA is not pure — it contains some RNA, proteins, and polysaccharides too.

CP14: Effect of Temperature on Enzyme Rate

Background Theory

Temperature affects enzyme activity through two competing mechanisms: (1) **increased kinetic energy** at higher temperatures means more frequent collisions between enzyme and substrate molecules, and more collisions have sufficient activation energy to form the enzyme-substrate complex (rate increases), (2) above the **optimum temperature**, the thermal energy breaks the weak bonds (hydrogen bonds, ionic bonds, hydrophobic interactions) maintaining the enzyme's tertiary structure, causing the active site to change shape (**denaturation**). Denaturation is irreversible — the substrate can no longer fit into the active site (induced-fit model fails). The result is a bell-shaped curve of rate vs temperature.

Apparatus

- Enzyme: amylase solution (for starch hydrolysis) or catalase source (for H_2O_2 decomposition)
- Substrate: 1% starch solution (for amylase) or H_2O_2 (for catalase)
- Water baths at 20, 30, 40, 50, 60°C (thermostatically controlled)
- For amylase: spotting tile, iodine solution (in potassium iodide), dropping pipette, stopwatch
- For catalase: gas syringe, conical flask with bung and delivery tube, stopwatch
- Thermometer, test tubes, syringes for measuring volumes

Method (Using Amylase and Starch)

1. Place tubes of **amylase** and **starch** solution in separate test tubes in the same water bath. **Equilibrate for 5 minutes** to ensure both solutions reach the target temperature before mixing.
2. Meanwhile, prepare a spotting tile with drops of iodine solution in each well.
3. Mix the amylase and starch. Start the stopwatch immediately.
4. Every **30 seconds**, remove a small sample with a pipette and add it to a well of iodine on the spotting tile. If starch is still present, the iodine turns **blue-black**. When all starch has been digested, the iodine remains **yellow-brown** (end-point).
5. Record the **time taken** for the iodine to no longer change colour (starch fully hydrolysed).
6. Repeat at each temperature (3 repeats). Calculate **rate = 1/time**.

Safety

- Care with hot water baths (burns risk). Use test tube tongs.
- Iodine solution stains skin and clothing — wear gloves and lab coat.

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- H_2O_2 is irritant — eye protection required.

Analysis

- Calculate rate = 1/time (s^{-1}) for each temperature.
- Plot **rate (y-axis) against temperature (x-axis)**. Expect a bell-shaped curve with a clear optimum.
- Below the optimum: rate increases with temperature (more kinetic energy → more enzyme-substrate collisions → more ES complexes formed per unit time). $Q_{10} \approx 2$.
- At the optimum: maximum rate; all factors balanced.
- Above the optimum: rate decreases sharply as the enzyme **denatures** — weak bonds in the tertiary structure break, the active site changes shape irreversibly, and the substrate can no longer bind.

Sources of Error and Improvements

Error	Effect	Improvement
Not pre-equilibrating enzyme and substrate at the target temperature	The reaction starts at the wrong temperature; initial rate is unreliable	**Always** equilibrate both tubes separately for at least 5 minutes in the water bath before mixing
Delay between mixing and starting the stopwatch	Misses the initial period of the reaction; underestimates rate	Start the stopwatch at the exact moment of mixing; have the stopwatch ready
Subjective colour judgement for end-point (iodine method)	Inconsistent determination of when starch is fully digested	Use a **colorimeter** set to a red filter (~580 nm) for objective measurement of starch-iodine colour; define a numerical threshold
Water bath temperature fluctuates	Actual temperature differs from the set temperature	Use thermostatically controlled water baths; check temperature with a thermometer before and after

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Exam Tip: Always equilibrate enzyme and substrate **separately** at the target temperature before mixing. This is critical — if you mix cold solutions and then heat them, the initial rate does not reflect the target temperature. Use **rate = 1/t** for valid comparisons. The optimum temperature for human enzymes is typically ~37°C; for thermophilic bacteria, it can be >80°C. Denaturation is **permanent** (irreversible) — this is different from reversible inhibition. Do not describe the enzyme as "killed" or "dead" — enzymes are not alive. Say "denatured."

CP15: Populations in a Habitat (Sampling Techniques)

Background Theory

Ecologists rarely count every organism in a habitat. Instead, they use **sampling techniques** to estimate population size and species diversity. The two key techniques are: (1) **quadrat sampling** for sessile (non-moving) organisms like plants, and (2) **mark-release-recapture (Lincoln index)** for mobile animals. **Random sampling** is essential to avoid bias — use random number tables or a random number generator to determine coordinates for quadrat placement. **Simpson's Diversity Index** quantifies biodiversity in a habitat, accounting for both species richness (number of species) and evenness (relative abundance of each species).

Apparatus

- Quadrats (e.g. 0.25 m^2 or 1 m^2), tape measures (for laying out a grid), random number table or calculator
- Pitfall traps, sweep nets, pooters, or pond nets (for mark-release-recapture of invertebrates)
- Non-toxic, non-harmful marker (waterproof paint pen, correction fluid, nail varnish — for marking animals)
- Identification keys, field guides, recording sheets, tally counter, calculator

Method

A. Quadrat Sampling (for plants and sessile organisms):

1. Lay out two tape measures at right angles along two edges of the study area to create a coordinate grid.
2. Generate random pairs of coordinates using a random number table.
3. Place the quadrat at each set of random coordinates.
4. For each quadrat: record the species present and either **count individuals** (for countable organisms), estimate **percentage cover** (for spreading plants like grasses, mosses), or use an **ACFOR scale** (Abundant, Common, Frequent, Occasional, Rare).
5. Use enough quadrats to give a representative sample (typically 10–30 depending on area size and habitat heterogeneity). The mean number of individuals per quadrat multiplied by the total number of quadrats that would fit in the area gives an estimate of total population.

B. Mark-Release-Recapture (Lincoln Index) (for mobile animals):

1. **Capture** a sample of individuals using an appropriate method (pitfall traps for ground beetles, sweep nets for meadow invertebrates). Count and record the number captured = **M**.
2. **Mark** each individual with a small, non-toxic, non-conspicuous mark (e.g. a dot of nail varnish on the wing case of a beetle). The mark must not affect survival, predation risk, or behaviour.
3. **Release** all marked individuals back into the habitat. Allow sufficient time for them to mix randomly with the unmarked population (at least 24 hours for most invertebrates).
4. **Recapture** a second sample from the same area. Count the total number in the second sample = **n**. Count the number of marked individuals in the second sample = **m**.
5. Estimate total population: **N = (M x n) / m** (Lincoln index formula).

Safety

- Complete a fieldwork risk assessment (weather, terrain, slippery surfaces, water bodies, stinging nettles, ticks).
- Handle all organisms gently and return them to their habitat promptly.
- Wash hands thoroughly after handling soil, water, or organisms.
- Wear appropriate clothing and footwear for the terrain.

Analysis

Lincoln Index: N = (M x n) / m

Assumptions (must be stated for full marks):

- The marks do not affect survival or make organisms more visible to predators
- The marks do not rub off or fade between sampling periods
- The population is **closed** (no immigration, emigration, births, or deaths between samples)
- Sufficient time has elapsed for marked individuals to mix randomly with unmarked individuals
- Sampling methods are the same for both capture events (equal probability of capture)

Simpson's Diversity Index:

> **$D = 1 - \Sigma(n/N)^2$**
> Where n = number of individuals of each species, N = total number of individuals of all species.
> D ranges from 0 (no diversity — single species) to 1 (infinite diversity). Higher D = greater biodiversity.

Sources of Error and Improvements

Error	Effect	Improvement
Marks wear off or are lost between sampling events	Fewer marked individuals found in recapture; m decreases; N is overestimated	Use permanent, waterproof markers; check mark durability beforehand
Marked animals more conspicuous to predators	Marked individuals are preferentially eaten; m decreases; N is overestimated	Use small, inconspicuous marks in locations not visible to predators
Immigration or emigration between sampling events	Population not closed; estimate becomes inaccurate	Sample within a short time frame; use a clearly bounded habitat; state the assumption
Non-random quadrat placement (e.g. avoiding muddy areas)	Biased species counts; unrepresentative of the true population	Use a random number table or random number generator for coordinates; accept whatever falls in the quadrat

Insufficient number of quadrats	Unrepresentative sample; high variance in estimates	Use at least 10–30 quadrats; increase number until the running mean stabilises

Exam Tip: The Lincoln index formula is **N = (M x n) / m**. Always state all assumptions. If marks affect survival, the formula gives an **overestimate** (fewer marked individuals survive to be recaptured, so m is low, inflating N). Simpson's Diversity Index: **$D = 1 - \Sigma(n/N)^2$**. A habitat with many species in roughly equal numbers has a high D (close to 1). A habitat dominated by one species has a low D (close to 0), even if there are many species present. You must be able to calculate D from given data.

CP16: Effect of an Environmental Factor on Plant Growth (Tropisms)

Background Theory

Plants respond to environmental stimuli by growing towards or away from them — these directional growth responses are called **tropisms**. They are mediated by the plant hormone **auxin (indole-3-acetic acid, IAA)**, which is produced in the shoot tip and distributed unevenly in response to the stimulus.

- **Phototropism:** Growth in response to light direction. In shoots, auxin moves to the **shaded side**, where it stimulates cell elongation. The shaded side elongates more than the lit side, causing the shoot to **bend towards the light** (positive phototropism). In roots, auxin inhibits cell elongation, so roots grow away from light (negative phototropism).
- **Gravitropism (geotropism):** Growth in response to gravity. In horizontally-oriented roots, auxin accumulates on the **lower side**, where it inhibits elongation — the upper side grows faster, curving the root **downwards** (positive gravitropism). In shoots, auxin on the lower side stimulates elongation — the lower side grows faster, curving the shoot **upwards** (negative gravitropism).

The differential response of roots and shoots to auxin concentration is explained by different **sensitivity thresholds**: shoots respond optimally at higher auxin concentrations, while roots are inhibited by the same concentrations.

Apparatus

- Seedlings (e.g. cress, mustard, oat coleoptiles), pots with soil or cotton wool
- Cardboard box with a single hole in one side (for unidirectional light), light source
- Clinostat (slowly rotating platform that eliminates gravity as a directional stimulus)
- Ruler, protractor, camera (for recording curvature over time), labels

Method

1. Grow seedlings in identical conditions until they are a few centimetres tall (typically 3–5 days for cress).
2. **Phototropism:** Place seedlings inside a light-proof box with a single hole in one side for unidirectional illumination. Place a lamp outside the hole. Leave for 24–48 hours. Record the angle of curvature towards the light daily using a protractor and/or photographs.
3. **Gravitropism:** Place a seedling (in its pot) on its side horizontally. After 24–48 hours, observe the direction of root and shoot growth. The shoot should curve upwards (negative gravitropism) and the root downwards (positive gravitropism).
4. **Control:** Place seedlings on a slowly rotating **clinostat**. Because the direction of gravity is constantly changing (rotating through 360°), there is no net directional gravitropic stimulus. Seedlings on the clinostat should grow straight (no curvature). This proves that the curvature in the static experiment is a genuine gravitropic response.
5. Measure **shoot/root length** and **angle of curvature** daily. Use multiple seedlings per treatment (at least 10).

Safety

- Wash hands after handling soil, compost, or seeds.
- Keep electrical equipment (lamps, clinostats) away from water.

Analysis

- Plot **mean angle of curvature** against time for each condition (unidirectional light, horizontal, clinostat).
- Shoots show **positive phototropism** (bend towards light) and **negative gravitropism** (grow upwards when placed horizontally).
- Roots show **positive gravitropism** (grow downwards).
- Clinostat-grown seedlings show no directional curvature (confirms gravity is the stimulus for gravitropism).
- Explain using auxin redistribution: auxin produced at shoot tip → moves to shaded/lower side → stimulates elongation in shoots (bends towards light/upwards) but inhibits elongation in roots (bends downwards).

Sources of Error and Improvements

Error	Effect	Improvement
Light leaking into the box from gaps or other sources	Light stimulus is not truly unidirectional; weak or absent phototropism	Seal the box thoroughly with black tape; use blackout material; work in a darkened room
Variable seedling ages and sizes	Different growth responses and rates	Germinate all seeds at the same time; select seedlings of similar size
Temperature differences between treatments	Confounding variable: temperature affects growth rate independently	Keep all experiments at the same temperature; avoid placing near heat sources
Measuring curvature angle subjectively (by eye)	Inaccurate or inconsistent data	Take photographs at the same angle; use image analysis software; use a protractor against a vertical reference line
Clinostat speed too fast or too slow	If too fast, centrifugal force replaces gravity; if too slow, gravity is not adequately nullified	Use a standard clinostat speed (1–4 rpm); follow manufacturer guidelines

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Exam Tip: Auxin (IAA) is produced in the shoot tip and redistributes in response to light and gravity. It **stimulates cell elongation in shoots** (positive phototropism: bends towards light; negative gravitropism: grows upwards) but **inhibits elongation in roots** at the same concentration (positive gravitropism: roots grow downwards). The **clinostat** eliminates gravity as a directional stimulus by continuous slow rotation — this is an important control. Always explain the mechanism: auxin moves to the shaded/lower side → causes differential cell elongation → curvature. This is an example of a hormone affecting gene expression (auxin activates genes encoding expansins and proton pumps that loosen the cell wall).